

⇒ What is AI:

AI is a computer software that can learn, think, and make decisions like a human.

Normally, computers can only do what humans explicitly code them to do.

But AI is trained using huge amounts of data so it can figure things out on its own.

Everyday ex. of AI.

- Streaming Recommendations:

Netflix or Youtube recommending videos you might like based on what you watched before.

- Voice Assistants: Siri or Google Assistant understanding your spoken accent and answering your questions.

Is AI a software?

Yes. AI is software.

It is not a physical robot, a metal body, or a mechanical brain. AI consists of programs, code, algorithms, and data that run on computers, smartphones, servers, or cloud systems.

ex. of AI software.

chatbot and Gmail. AI software programs that read, understand, and write human text.

What is ML?

ML is the method used to teach computers how to learn from experience (data) without being step-by-step programmed.

Imagine teaching a child to recognize a dog.

o Traditional Approach (old coding):

You give the child strict rules:

"A dog has four legs, fur, two ears and barks"
(if they see a CAT with four legs and fur, they might get confused)

o Machine Learning Approach: you show the child 1,000 photos of dogs and say, "This is a dog." Over time, the child's brain automatically spots patterns (snout shape, tail movement, body shape). Soon, they can recognize any dog, even a breed they have never seen before.

This is ML: You feed the computer data and it learns the pattern on its own.

→ How is ML Different from AI?

- AI: is the Goals. Making machines smart enough to mimic human intelligence.

- ML: is the Method: The specific technique used to achieve that goal.

Think of AI as a vehicle and ML as the Engine. Every motorcar is a vehicle, but the engine is the specific technology inside that makes it run.

Difference
Feature

AI

ML

Definition

> The broad concept of machines performing tasks. Mostly of AI where machines

Scope

Very broad (includes logic, rule-based systems, and learning).

Goal

To mimic human intelligence and decision making

A specific subset of AI where machines learn from data.

Narrower (focuses strictly on data, pattern and Algorithms.)

To learn from past data to make accurate predictions

How it works, Can use hand-coded rules or learning systems.

Must use data to train itself over time

ex: Spam Email Filter:

- Old AP (Rule based): A programmer writes a rule, "if an email contains the word 'FREE Money', move it to spam." (Spammers easily bypass this by writing "F-R-E-E M-O-N-E-Y")

- ML-based AP: The computer analyzes millions of spam and non-spam emails, it notices subtle combinations of words, sending times, and sender addresses, it automatically updates its understanding to block spam, even when spammers change their tricks

⇒ Types of ML:

1. Supervised Learning:
2. Unsupervised Learning.
3. Reinforcement Learning.

1. Supervised Learning:

The computer is given labeled data, this means you provide both the input question AND the correct answer (label), the algorithm learns the pattern b/w the two so it can give the answer for new, unseen inputs.

ex: Student Exam predictions

Study hours per week (x_1)	Attendance Percentage (x_2)	Practice Tests Taken (x_3)
12	90%	5
3	55%	1
8	80%	4
2	60%	2

How supervised learning works here?

1. The 3. Features (X):

- X_1 = Study hours.
- X_2 = Attendance %
- X_3 = Practice Tests

2. The label (Y):

- 1 = Pass, 0 = Fail

3. Learning the pattern: The algorithm process these rules and spots a rule, such as
"If a student studies more than 5 hours AND has above 75% attendance, they usually pass (1)." ^{input}

4. Teaching a New Student (output)

Exam outcome (Y-label)

1 (Pass)

0 (Fail)

1 (Pass)

0 (Fail)

You enter a new student's record: [10 hours, 85% attendance, 3 tests].

The model process these 3 numbers and predicts: 1 (Pass)

2, Unsupervised Learning ("Learning on your own")

Def: The computer is given unlabeled (data without any correct answer). The algorithm's job is to explore the data and find hidden patterns, clusters, or groupings on its own.

ex:

Dataset Online store ke customers: from e-commerce website 4 customers ko data collect krke hair dene ki answer/label ke.

Monthly visits (x_1)	Items Viewed (x_2)	Money spent in \$ (x_3)
25	120	\$ 500
2	50	\$ 20
30	150	\$ 800
3	80	\$ 15

Note: Here we don't have payoff or "Correct label". There is no column of answer.

So how algorithm works?

1. Numbers comparison (Distance check)
Algorithm compare all numbers with each other.

o Customer 1 ([25, 125, 500]) and customer 3 ([30, 150, 800]) ke numbers

are larger and they are nearby with each other!

2/ Groups (clusters) creating
it will find patterns by their own,
Algorithm creates 2 groups:

- o Group A (High Activity): customer 1 and customer 3
- o Group B (Low Activity): customer 2 and customer 4

3/ (Business Decision)s

Algorithm just creates groups, then humans assign the name to that Group!

ex.

- o Group A = "vip shoppers"
- o Group B = casual browsers

Group/Label
?
?
?
?
?

4 Reinforcement Learning ("Learning through Rewards and Penalties")

Def: There is no dataset before hand, instead, an AI "Agent" learns by interacting with an environment through trial and error.

- if it makes a good move $\rightarrow$ it gets Reward
- if it makes a bad move $\rightarrow$ it gets a Penalty

Over millions of attempts, the AI learn the best strategy to maximize its total reward.

ex: Training a puppy. When the puppy sits on command, you give it a treat (Reward) when it bites the sofa, you say "No!" (Penalty)

Eventually, the puppy learns how to behave to get the maximum number of treats.

No static dataset (Trial and Error)

3, Semi-Supervised Learning ("learning with a little help")

Def: A hybrid of supervised and unsupervised learning. The computer is given a small set of labeled data and large amount of unlabeled data.

Labeling data manually takes huge time and money. Semi-supervised learning uses the small labeled set to get a baseline understanding, then uses the unlabeled data to scale up its learning.

ex: A teacher solves 2 ^{practice} problems on the whiteboard and then hands the student 100 practice problems to figure out using the same basic principles.